Rigid Analytic Geometry Week 6 Solution

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Problem 1

That the Grothendieck topology $\mathfrak{I}$ satisfies that for any $\mathcal{S} \in \mathfrak{I}_U$,

$$U = \bigcup_{V \in \mathcal{S}} V$$

by Problem 9 in the previous sheet thus it is a G_+-space. The Grothendieck topology forcing $\mathfrak{B}$ to be a set of quasicompact sets is proven in the lecture. Remains to show that it is a G_+-space.

Let $\mathcal{T} = [V \mid V \in \mathfrak{B} \cap \mathfrak{Q}_C(U)]$ and $V \in \mathfrak{B} \cap \mathfrak{Q}_C(U)$. Since $\mathfrak{B}$ is a basis of a topology, $\mathcal{T}|_V$ covers V. By GT-Loc, $\mathcal{T} \vDash U$ that is it covers U.

Problem 3

Let us denote the new condition as V4'. For V4$\Rightarrow$V4', if $[\Theta_1, \cdots, \Theta_n]$ covers U, thus there is $\Theta \in [\Theta_1, \cdots, \Theta_n] \cap \xi_{\mathfrak{B}}$. By definition there is i such that $\Theta \subseteq \Theta_i$. By V3, $\Theta_i \in \xi_{\mathfrak{B}}$.

Conversely, let $\Omega \in \xi_{\mathfrak{B}}$ and $\mathcal{S} \subseteq \mathfrak{B}_\Omega$ be a covering Ω-sieve. Then by quasicompactness of Ω, there is $\Theta_1, \cdots, \Theta_n$ in $\mathcal{S}$ covering U. By V4', Θ_i is in $\xi_{\mathfrak{B}}$. Therefore $\Theta_i \in \mathcal{S} \cap \xi_{\mathfrak{B}}$.